

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	58987	52.261471
1	[Git, Change, Create]	17329	15.353197
2	[Git, Commit]	11316	10.025782
3	[Git, Change, Delete]	8787	7.785131
4	[File, Git]	5921	5.245905
5	[Git, Change, Rename]	3354	2.971587
6	[Git, Tag]	1795	1.590339
7	[Author, Git, Person]	1268	1.123426
8	[Json, Key]	668	0.591837
9	[Json, Value, Scalar]	603	0.534248
10	[Committer, Git, Person]	370	0.327814
11	[NPM, Dependency]	338	0.299462
12	[Type, TS, Primitive]	293	0.259593
13	[Type, TS, Declared]	277	0.245417
14	[TS, ExternalDeclaration]	215	0.190486
15	[Git, Change, Copy]	138	0.122266
16	[Type, TS, Literal]	136	0.120494
17	[Json, Value, Object]	133	0.117836
18	[Type, TS, Union]	120	0.106318
19	[Type, TS, ObjectMember]	103	0.091256
20	[NPM, Script]	91	0.080624
21	[TS, Property]	65	0.057589
22	[TS, Function]	47	0.041641
23	[Git, Branch]	46	0.040755
24	[Type, TS, FunctionParameter]	40	0.035439
25	[Type, Object, TS]	39	0.034553
26	[File, Directory]	34	0.030123
27	[Type, TS, Function]	34	0.030123
28	[TS, Parameter]	33	0.029237
29	[Package, File, Json, NPM]	29	0.025694

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

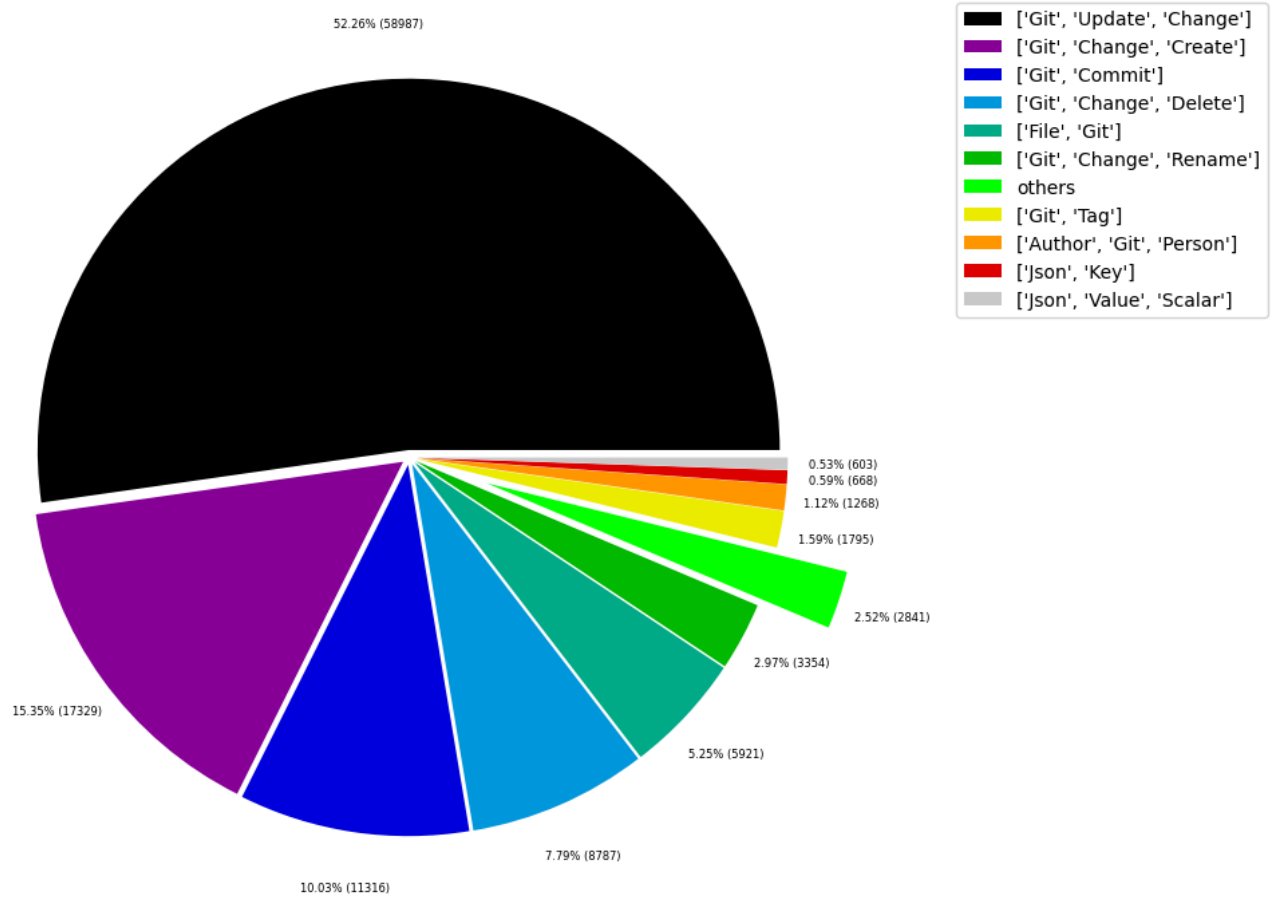


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, JQAssistant]	1	0.000886
1	[NPM, PublishConfig]	1	0.000886
2	[File, TS, Scan]	1	0.000886
3	[TS, Method]	1	0.000886
4	[Repository, File, Git]	1	0.000886
5	[TS, Constructor]	1	0.000886
6	[Value, TS, ObjectMember]	1	0.000886
7	[TS, Class]	1	0.000886
8	[TS, Enum]	2	0.001772
9	[Value, Object, TS]	3	0.002658
10	[Type, TS, Tuple]	3	0.002658
11	[Value, TS, Function]	4	0.003544
12	[TS, TypeParameter]	4	0.003544
13	[Value, TS, Complex]	5	0.004430
14	[Repository, NPM]	5	0.004430
15	[NPM, Engine]	6	0.005316
16	[Project, TS]	6	0.005316
17	[File, Local]	6	0.005316
18	[Type, TS, TypeParameterReference]	6	0.005316
19	[Value, TS, Member]	6	0.005316
20	[File, TS, Local, Module]	6	0.005316
21	[Value, TS, Call]	6	0.005316
22	[TS, EnumMember]	8	0.007088
23	[Type, TS, NotIdentified]	11	0.009746
24	[TS, ExternalModule]	11	0.009746
25	[Json, Value, Array]	12	0.010632
26	[Value, TS, Declared]	13	0.011518
27	[TS, TypeAlias]	16	0.014176
28	[File, Directory, Local]	16	0.014176
29	[TS, Interface]	17	0.015062

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

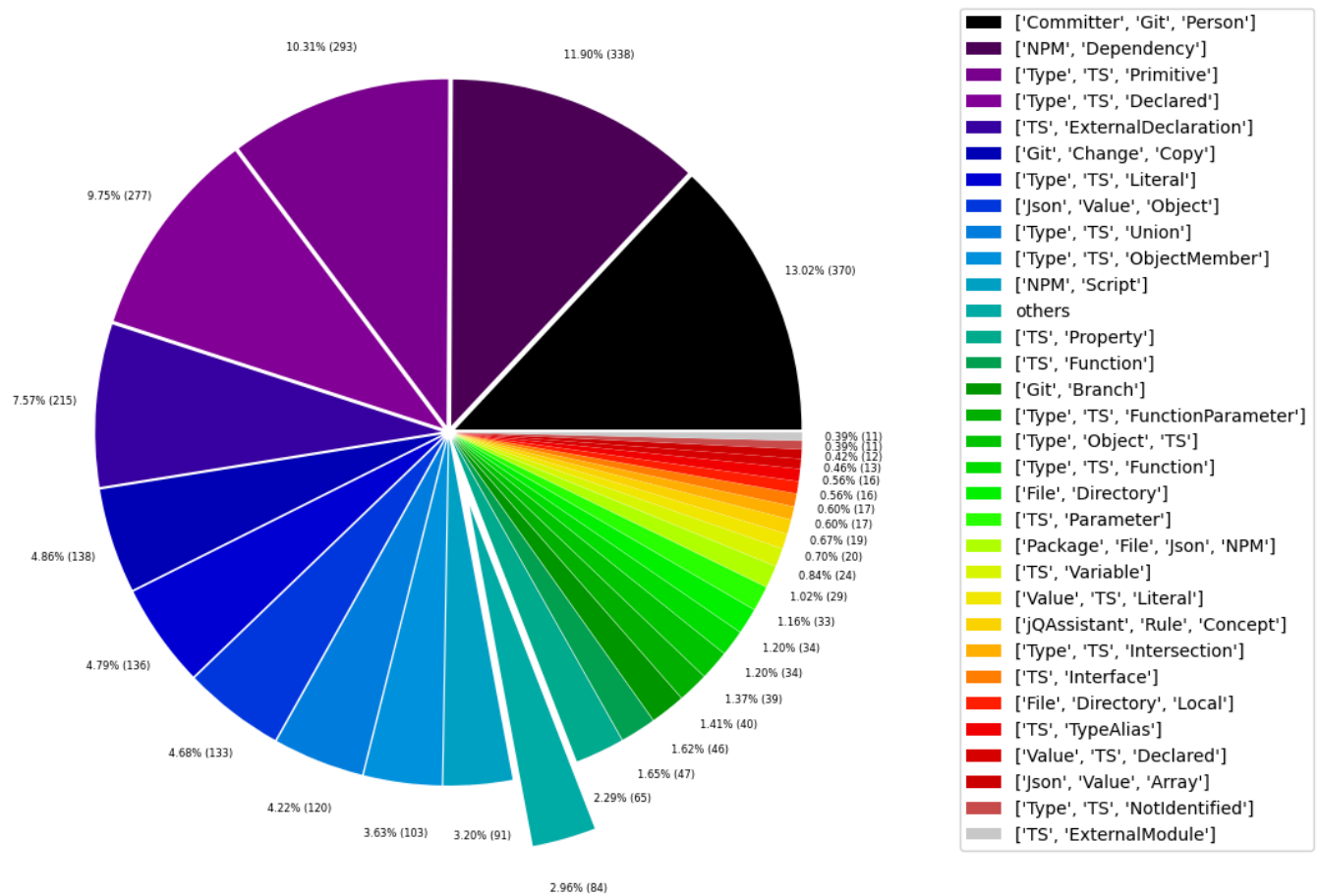


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

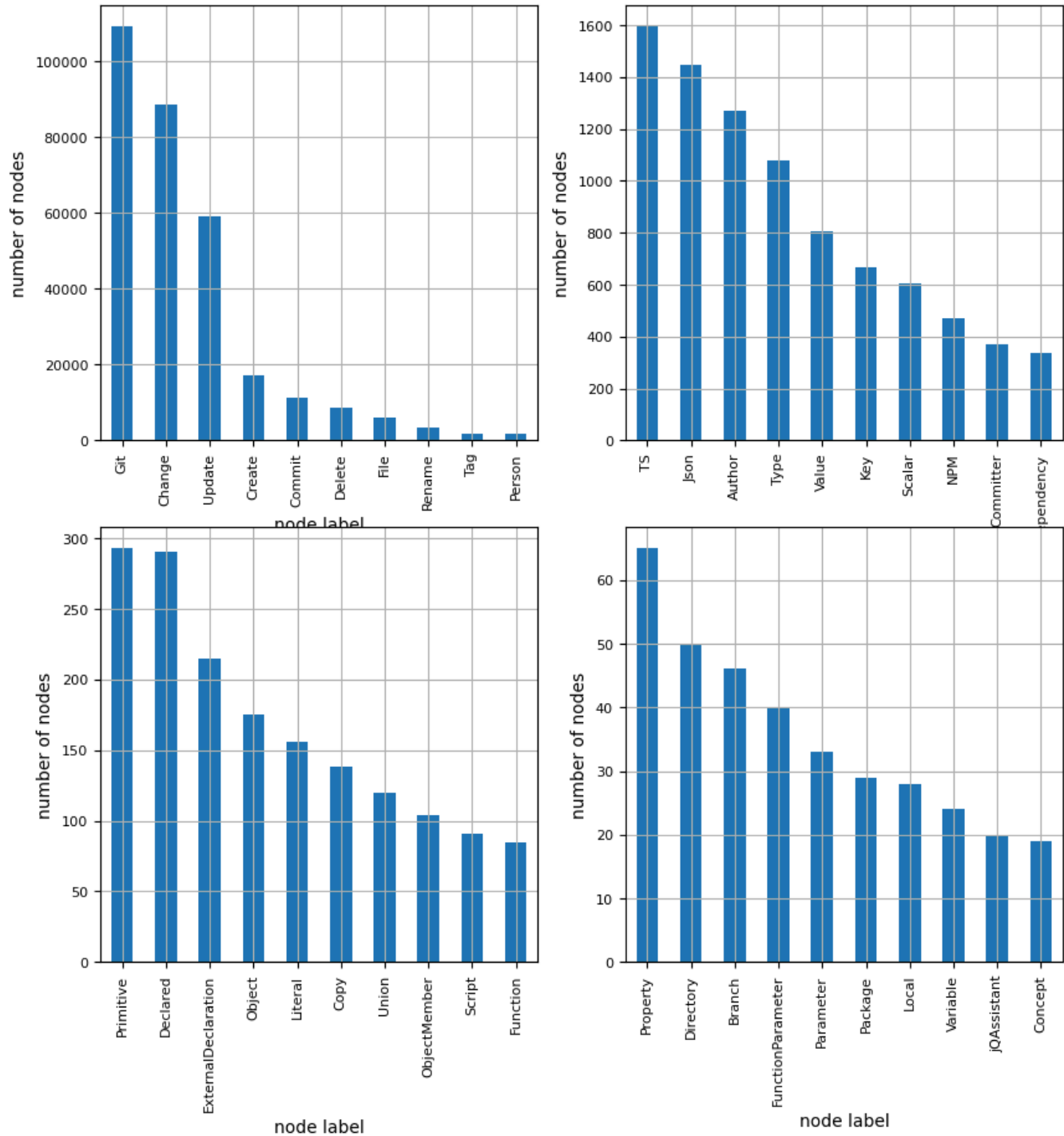
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	109312	96.848559
1	Change	88595	78.493652
2	Update	58987	52.261471
3	Create	17329	15.353197
4	Commit	11316	10.025782
5	Delete	8787	7.785131
6	File	6014	5.328301
7	Rename	3354	2.971587
8	Tag	1795	1.590339
9	Person	1638	1.451240
10	TS	1595	1.413143
11	Json	1445	1.280245
12	Author	1268	1.123426
13	Type	1079	0.955976
14	Value	806	0.714102
15	Key	668	0.591837
16	Scalar	603	0.534248
17	NPM	470	0.416412
18	Committer	370	0.327814
19	Dependency	338	0.299462
20	Primitive	293	0.259593
21	Declared	290	0.256935
22	ExternalDeclaration	215	0.190486
23	Object	175	0.155047
24	Literal	156	0.138213
25	Copy	138	0.122266
26	Union	120	0.106318
27	ObjectMember	104	0.092142
28	Script	91	0.080624
29	Function	85	0.075309
30	Property	65	0.057589
31	Directory	50	0.044299
32	Branch	46	0.040755
33	FunctionParameter	40	0.035439
34	Parameter	33	0.029237
35	Package	29	0.025694
36	Local	28	0.024808
37	Variable	24	0.021264
38	jqAssistant	20	0.017720
39	Concept	19	0.016834

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 338894

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	88595	26.142393
1	MODIFIES	88595	26.142393
2	UPDATES	58987	17.405737
3	COMMITTED	22632	6.678194
4	CREATES	20821	6.143809
5	HAS_PARENT	12402	3.659551
6	DELETES	12141	3.582536
7	HAS_COMMIT	11316	3.339097
8	HAS_FILE	5921	1.747154
9	RENAMES	3354	0.989690
10	CHANGED_TOGETHER_WITH	1910	0.563598
11	HAS_TAG	1795	0.529664
12	ON_COMMIT	1795	0.529664
13	HAS_NEW_NAME	1770	0.522287
14	HAS_AUTHOR	1268	0.374158
15	DEPENDS_ON	887	0.261734
16	HAS_KEY	668	0.197112
17	HAS_VALUE	668	0.197112
18	CONTAINS	596	0.175866
19	HAS_COMMITTER	370	0.109179
20	OF_TYPE	339	0.100031
21	EXPORTS	309	0.091179
22	REFERENCES	197	0.058130
23	DECLARES	186	0.054884
24	DECLARES_DEV_DEPENDENCY	169	0.049868
25	DECLARES_DEPENDENCY	161	0.047507
26	COPIES	138	0.040721
27	HAS_MEMBER	104	0.030688
28	HAS_TYPE_ARGUMENT	94	0.027737
29	DECLARES_SCRIPT	91	0.026852

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

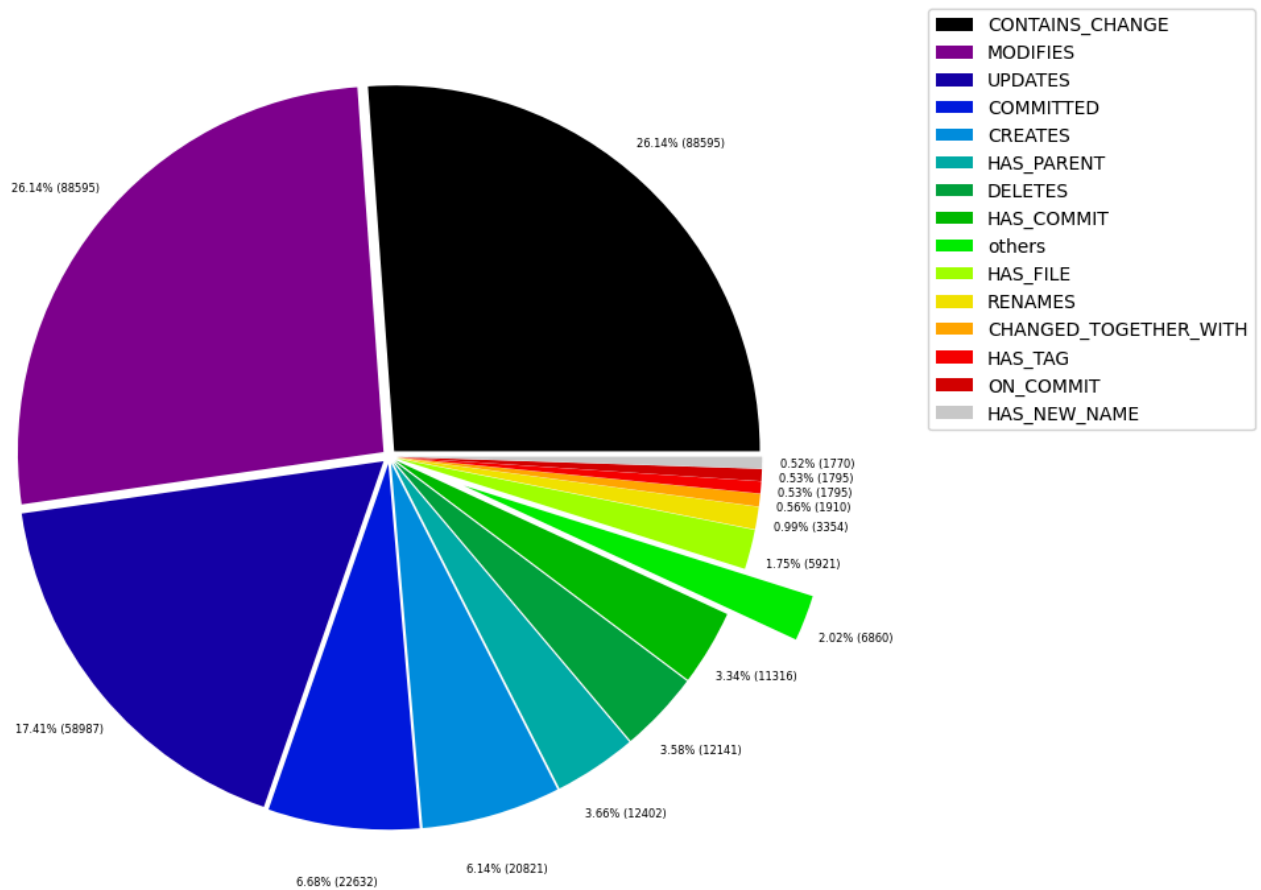


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

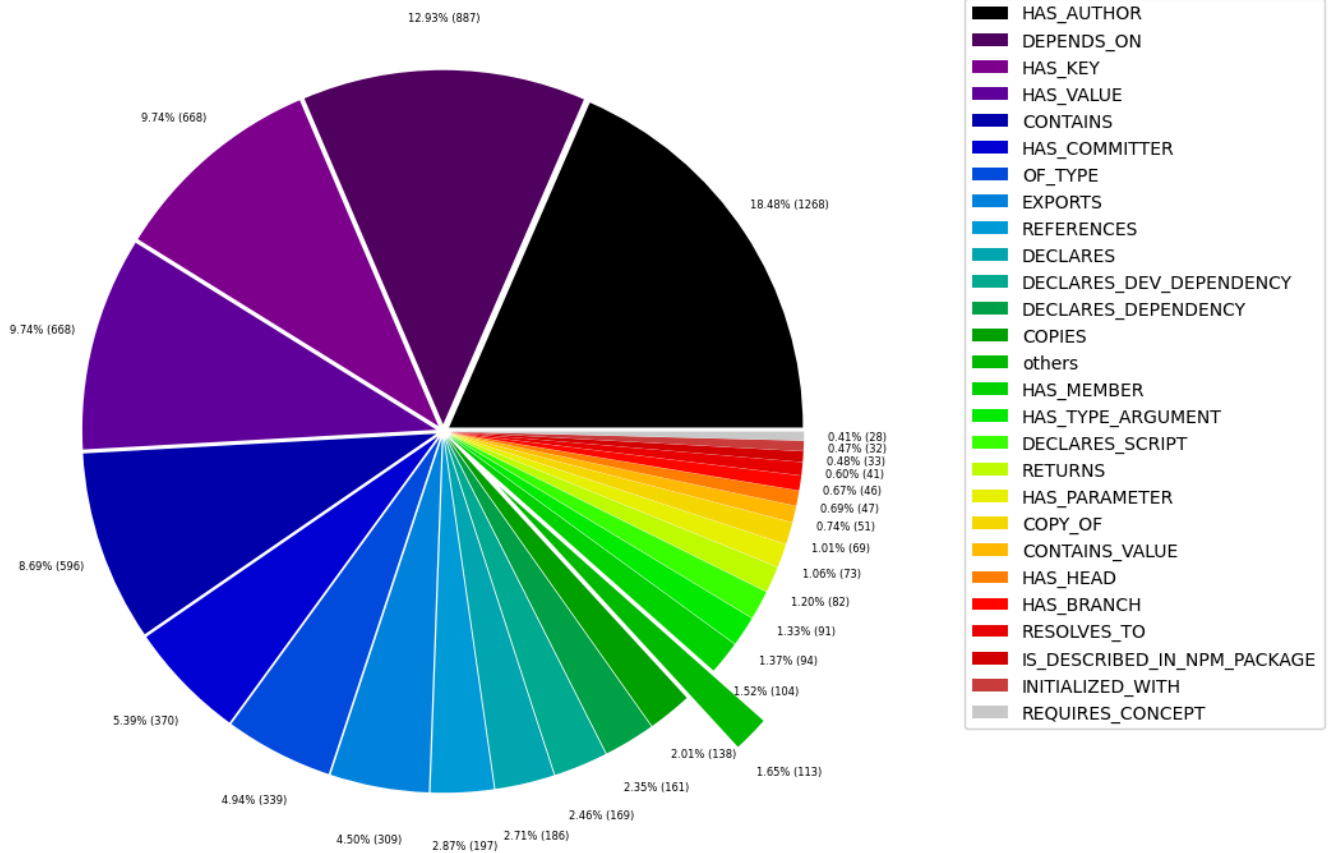
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	DECLARES_PUBLISH_CONFIG	1	0.000295
1	CONSTRAINED_BY	4	0.001180
2	IN_REPOSITORY	5	0.001475
3	REFERENCED_PROJECTS	5	0.001475
4	PARENT	6	0.001770
5	MEMBER	6	0.001770
6	HAS_ROOT	6	0.001770
7	HAS_NPM_PACKAGE	6	0.001770
8	HAS_CONFIG	6	0.001770
9	HAS_ARGUMENT	6	0.001770
10	EXTENDS	6	0.001770
11	DECLARES_ENGINE	6	0.001770
12	CONTAINS_PROJECT	6	0.001770
13	CALLS	6	0.001770
14	DECLARES_PEER_DEPENDENCY	8	0.002361
15	USES	11	0.003246
16	INCLUDES_CONCEPT	19	0.005606
17	REQUIRES_CONCEPT	28	0.008262
18	INITIALIZED_WITH	32	0.009442
19	IS_DESCRIBED_IN_NPM_PACKAGE	33	0.009738
20	RESOLVES_TO	41	0.012098
21	HAS_BRANCH	46	0.013574
22	HAS_HEAD	47	0.013869
23	CONTAINS_VALUE	51	0.015049
24	COPY_OF	69	0.020360
25	HAS_PARAMETER	73	0.021541
26	RETURNS	82	0.024196
27	DECLARES_SCRIPT	91	0.026852
28	HAS_TYPE_ARGUMENT	94	0.027737
29	HAS_MEMBER	104	0.030688

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberOfNodesWithSameLabelsAsTarget
0	[Git, Update, Change]	MODIFIES	[File, Git]	58987	58987	
1	[Git, Update, Change]	UPDATES	[File, Git]	58987	58987	
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	58987	11316	
3	[Git, Change, Create]	MODIFIES	[File, Git]	17329	17329	
4	[Git, Change, Create]	CREATES	[File, Git]	17329	17329	
5	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	17329	11316	
6	[Git, Commit]	HAS_PARENT	[Git, Commit]	12402	11316	
7	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	11316	1	
8	[Author, Git, Person]	COMMITTED	[Git, Commit]	11316	1268	
9	[Committer, Git, Person]	COMMITTED	[Git, Commit]	11316	370	
10	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	8787	11316	
11	[Git, Change, Delete]	MODIFIES	[File, Git]	8787	8787	
12	[Git, Change, Delete]	DELETES	[File, Git]	8787	8787	
13	[Repository, File, Git]	HAS_FILE	[File, Git]	5921	1	
14	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	3354	11316	
15	[Git, Change, Rename]	DELETES	[File, Git]	3354	3354	
16	[Git, Change, Rename]	RENAMES	[File, Git]	3354	3354	
17	[Git, Change, Rename]	MODIFIES	[File, Git]	3354	3354	
18	[Git, Change, Rename]	CREATES	[File, Git]	3354	3354	
19	[Repository, File, Git]	HAS_TAG	[Git, Tag]	1795	1	
20	[Git, Tag]	ON_COMMIT	[Git, Commit]	1795	1795	
21	[File, Git]	HAS_NEW_NAME	[File, Git]	1770	5921	
22	[File, Git]	CHANGED_TOGETHER_WITH	[File, Git]	1655	5921	
23	[Repository, File, Git]	HAS_AUTHOR	[Author, Git, Person]	1268	1	
24	[Json, Value, Object]	HAS_KEY	[Json, Key]	668	133	
25	[Json, Key]	HAS_VALUE	[Json, Value, Scalar]	552	668	
26	[Repository, File, Git]	HAS_COMMITTER	[Committer, Git, Person]	370	1	
27	[TS, Function]	DEPENDS_ON	[TS, ExternalDeclaration]	293	47	
28	[Package, File, Json, NPM]	CHANGED_TOGETHER_WITH	[Package, File, Json, NPM]	253	29	
29	[File, TS, Local, Module]	DEPENDS_ON	[TS, ExternalDeclaration]	236	6	

Graph Density

total_number_of_nodes (vertices): 112869

total_number_of_relationships (edges): 338894

-> total directed graph density: 2.660225015790846e-05

-> total directed graph density in percent: 0.002660225015790846